(II-58) Problem 25:

1. Show that the language $L = \{a^n b^n / n \geq 1\}$ is not regular

Soln:

Step 1: Assume the given Language 'L' is a regular language

Step 2: Calculate the number of states in DFA.

No. of states of $L = \{a^n b^n \mid n \geq 1\}$ is $n+n = 2n$

Step 3: Take one string 'w' and calculate the length of string

String, $w = a^n b^n$

Length of String $\}$ $|w| = n+n = 2n$

Step 4: If the length of string, $|w| \geq n$

i.e. $2n \geq n$

So, we can break into three strings.

Step 5: $w = xyz$. Let $w = a^n b^n$

Assume that $w = a^n b^n$

$xy = a^m$; $y = a^j$; $z = a^{n-m} b^n$

To check assumptions:

$xyz = a^m a^{n-m} b^n = a^n b^n$

So, our assumptions is correct.

(i) $|xy| \leq n$, $|a^m| \leq n \Rightarrow m \leq n$

(ii) $y \neq \epsilon$ (or) $|y| \geq 1$

$|a^j| \geq 1 \Rightarrow j \geq 1$

(iii) For all $k \geq 0$, the string xy^kz is also in 'L'

$xy^kz = xyy^{k-1}z$

$= a^m (a^j)^{k-1} a^{n-m} b^n = a^{m+j(k-1)+n-m} b^n = a^{n+j(k\ldots}$

$xyz = a \quad b^n$

Apply, $K=0$, $xy^Kz = a^{n+j(k-1)} b^n = a^{n-j} b^n \neq a^n b^n$

$K=1$, $xy^Kz = a^{n+j(k-1)} b^n = a^n b^n$

$K=2$, $xy^Kz = a^{n+j(k-1)} b^n \neq a^{n+j} b^n$

Since, for $K=0,2$ we have the string that does not belong to the language L, So the language

$L = \{a^n b^n \mid n \geq 1\}$ is not regular.

Problem 26: Prove that $L = \{a^n \mid n = i^2, i \geq 1\}$ (or)

$L = \{0^n \mid n = i^2, i \geq 1\}$ is not regular.

Soln:

Step 1: Assume that given language is a regular language

Step 2: Calculate the number of states

No. of States = 'n'

Step 3: Take any string 'w'

Let $w = a^n$, where $n = i^2$

The length of string $|w| = n$

Step 4: $|w| \geq n$

Therefore, $n \geq n$, We can break into 3 strings

Step 5: $W = a^n$

Let $w = xyz$

$xy = a^m$; $y = a^j$; $z = a^{n-m}$

(II-60) To check the ...

$xyz = a^m . a^{n-m} = a^n$

So, our assumption is correct.

(i) $|xy| \leq n$,

$|a^m| \leq n \Rightarrow m \leq n$

(ii) $|y| \geq 1$ (or) $y \neq \epsilon$

$|a^j| \geq 1 \Rightarrow j \geq 1$

(iii) For all $K \geq 0$, check the string xy^kz

$$xy^kz = xy\,y^{(k-1)}z$$
$$= a^m a^{j(k-1)} a^{n-m}$$
$$= a^{m+j(k-1)+n-m}$$
$$xy^kz = a^{n+j(k-1)}$$

$K=0$, $xy^0z = a^{n+j(0-1)} = a^{n-j} \neq a^n$ (false)

$K=1$, $xy^1z = a^{n+j(1-1)} = a^n = a^n$ (True)

$K=2$, $xy^2z = a^{n+j(2-1)} = a^{n+j} \neq a^n$ (false)

For $K = 0, 2$, We have the string that does not be to the language L, So the language

$L = \{a^n \mid n = i^2, i \geq 1\}$ is not regular language.

Problem 27: Show that $L = \{0^n 1 0^n \mid n \geq 1\}$ is not regular

Soln:

Step 1: Assume that given language is a regular language

No. of states in $\{0^n 1 0^n\}$ is $= n+1+n = 2n+1$

Step3: ~~Calculate the~~ Take a string 'w'. Calculate the length of w

$w = 0^n 1 0^n$

$|w| = 2n+1$

Step4: if length of string $\geq n$ (i.e.) $|w| \geq n$

$|w| = 2n+1 \geq n$

We can break into three strings.

Step5: $w = xyz$. Let $w = 0^n 1 0^n$

Assume $xy = 0^m$; $y = 0^j$; $z = 0^{n-m} 0^n . 1$

To check assumptions:

$xyz = 0^m 0^{n-m} 0^n . 1 = 0^{m+n-m} 0^n 1 = 0^n 1 0^n$

Our Assumptions are Correct.

(i) $|xy| \leq n$

$|0^m| \leq n \Rightarrow m \leq n$

(ii) $|y| \geq 1$

$|0^j| \geq 1 \Rightarrow j \geq 1$

(iii) For all $k \geq 0$, Check the string $xy^k z$

$xy^k z = xyy^{k-1} z$

$xy^k z = 0^m 0^{j(k-1)} 0^{n-m} 0^n . 1$

When $k=0$, $xy^0 z = 0^m 0^{j(0-1)} 0^{n-m} 0^n . 1 = 0^n 0^{j(-1)} 0^n . 1$

$\neq 0^n 0^n . 1$ (false)

When $k=1$, $xy^1 z = 0^m 0^{j(1-1)} 0^{n-m} 0^n . 1 = 0^n . 1 . 0^n$ (true)

When $k=2$, $xy^2 z = 0^m 0^{j(2-1)} 0^{n-m} 0^n . 1 \neq 0^n 1 0^n$ (false)

For $k = 0, 2$, we have string that does not belong to the lang. $L = \{0^n 1 0^n / n \geq 1\}$. So, L is not regular.

II-62)

Problem 28: Show that $L = \{ a^{i^2} \mid i \geq 1 \}$ is not regular.

Soln:

Step 1: Assume the language, L is regular.

Step 2: Find number of states in L

No. of states $= n^2$

Step 3: Find the length of string, W

$W = a^{i^2}$, $|W| = n^2$

$2n \geq n$

Step 4: Check whether $|W| \geq n^2$

$n^2 \geq n$, So break it into three strings.

Step 5: $W = xyz$

$xy = a^m$, $y = a^j$, $z = a^{i^2 - m}$

Check whether our assumption is correct

$xyz = a^m a^{i^2 - m} = a^{i^2}$

Our assumption is correct.

Step 6: (i) $|xy| \leq n$

$|a^m| \leq n \Rightarrow |m| \leq n$

(ii) $|y| \geq 1$

$|a^j| \geq 1 \Rightarrow j \geq 1$

(iii) for $k \geq 0$, Check ~~whether~~ for the string xy^kz

$xy^kz = xyy^{k-1}z = a^m a^{j(k-1)} a^{i^2 - m}$

for $k = 0$, $xy^0z = a^m a^{j(0-1)} a^{i^2 - m} \neq a^{i^2}$ (false)

$k = 1$, $xy^1z = a^m a^{j(1-1)} a^{i^2 - m} = a^{i^2}$ (true)

$k = 2$, $xy^2z = a^m a^{j(2-1)} a^{i^2 - m} \neq a^{i^2}$ (false)

For $K = 0, 2$, the string doesnot belong to the language, L. So, $L = \{ a^{i^2} \mid i \geq 1 \}$ is not regular.

Problem ... $\{0^n \mid \ldots \text{ square}\}$ is not regular language.

Soln:

The language L Can be represented as, $L = \{0^n \mid n = n^2\}$

Step1: Assume that language L is a regular language.

Step2: Calculate No. of States.

$$\text{No. of states} = n^2$$

Step3: Take string, w and Find length of w, $|w|$

$$w = 0^n \; ; \; |w| = n^2$$

Step4: If $|w| \geq n$, then divide into three strings.

$$|w| \geq n \Rightarrow n^2 \geq n$$

Break it into three strings.

Step5:

$$xy = a^m \; ; \; y = a^j \; ; \; z = a^{n-m}$$

$$xyz = 0^m \cdot 0^{n-m} = 0^n$$

Our Assumption is Correct.

$$xy^kz = xyy^{k-1}z = 0^m\, 0^{j(k-1)}\, 0^{n-m}$$

for $k=0$, $xy^0z = 0^m\, 0^{j(0-1)}\, 0^{n-m} \neq 0^n$

for $k=1$, $xy^1z = 0^m\, 0^{j(1-1)}\, 0^{n-m} = 0^n$

for $k=2$, $xy^2z = 0^m\, 0^{j(2-1)}\, 0^{n-m} \neq 0^n$

for $k=0,2$, the string doesnot belongs to the Regular language.

$\therefore$ $L = \{0^n \mid n \text{ is a perfect square}\}$ is not regular.

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(II-64) **Problem 30:**

Prove that set of strings of 0's and 1's that are of the form WW^R, that is some string followed by its reverse, such that $L = \{WW^R / w \in (0,1)\}$

Soln: Let $W = 0^n 1^n \Rightarrow W^R = 1^n 0^n$

The string is $= 0^n 1^n 1^n 0^n$

$$L = \{0^n 1^n 1^n 0^n \mid n \geq 1\}$$

Step 1: Assume the language, L is regular language.

Step 2: No. of states $= n+n+n+n = 4n$

Step 3: $|W| = 4n$

Step 4: $|W| \geq n$; $4n \geq n$

break it into three strings.

Step 5: $xy = 0^m$; $y = 0^j$; $z = 0^{n-m} 1^n 1^n 0^n$

$xyz = 0^m 0^{n-m} 1^n 1^n 0^n = 0^n 1^n 1^n 0^n$

Our Assumption is correct.

$xy^k z = xyy^{k-1}z = 0^m 0^{j(k-1)} 0^{n-m} 1^n 1^n 0^n$

If $K=0$, $xy^0 z = 0^m 0^{j(0-1)} 0^{n-m} 1^n 1^n 0^n \neq 0^n 1^n 1^n 0^n$

If $K=1$, $xy^1 z = 0^m 0^{j(1-1)} 0^{n-m} 1^n 1^n 0^n = 0^n 1^n 1^n 0^n$

If $K=2$, $xy^2 z = 0^m 0^{j(2-1)} 0^{n-m} 1^n 1^n 0^n \neq 0^n 1^n 1^n 0^n$

For $K = 0, 2$, the string doesnot belong to the regular language.

$\therefore L = \{WW^R / w \in (0,1)\}$ is not regular

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